

Assignment 1

September 25, 2024

Question 1

1a) Treatment Variable X: pescatarian diet Outcome Variable Y: Alzheimer's disease Possible confounder Z: Age Explanation: Younger generations are more likely to take on a pescatarian or vegetarian diet due to the positive effect on the climate and the earth. Older generations are more susceptible to getting Alzheimer's disease. In short, age could have influence on both diet and the risk of getting Alzheimer's.

1b) Treatment Variable X: Dentist visit Outcome Variable Y: Number of cavities Possible confounder Z: Dietary intake Explanation: Frequent dentist visit usually have small effect on one's preference for food. Certain types of food, especially those with high sugar content, are more likely to cause the formation of cavities.

1c) Treatment Variable X: Vending machine Outcome Variable Y: Diabetes Possible Confounder Z: Family income level Explanation: Lower family income levels are more likely to enroll in public schools that have more access to vending machines. Family income level also determines awareness of a healthier lifestyle and access to healthy food.

1d) Treatment Variable X: Sauna use Outcome variable Y: Life span Possible confounder Z: Income level Explanation: A person with a higher income level is more likely to have access to a sauna. A person with a higher income level can also afford a healthier lifestyle (Better food, shelter, and living habits) which prolongs life span.

Question 2

2a) Units of analysis: Individual (First year college students) Explanatory variable: Sleep Outcome variable: GPA

2b) The subjects are the first-year college students. The subjects belong to different groups since all the data are drawn from three different universities and there are subgroups like genders and different socioeconomic backgrounds. Variables measured for each subject are their total sleep time in the early term.

2c) The study is observational. It would require a lot of effort to change one's sleep schedule forcefully, especially if the study tracks sleep across a period of time. Moreover, if GPA suggests future job success and sleep is hypothesized to be one of the influence variables, it would be unethical to force students to sleep for a certain time just to study the impact of sleep on GPA.

2d) One of the previous studies collected students' sleeping data from self-reporting, which can be significantly biased. Some possible biases include social responsiveness bias and obsequiousness bias. Student may report inaccurate time because they want to present a better version of themselves in front of others, and also it would be hard to keep track of the accurate hours everyday.

2e) One of the confounder the study tries to eliminate is daytime sleep. One of the confounder the study did not try to explain is substance use.

2f) 6 hours

Question 3 3a)

```
[6]: import numpy as np

def flip(p):
    if (p < 0 and p > 1):
        raise ValueError("Enter a value between 0 and 1(Not including 1 and 0)")
    random = np.random.random()
    return random <= p

p=0.26
for i in range(4):
    print(flip(p))
```

True

True

False

False

I expect 1 head out of the four toss when $p = 0.26$

3b)

```
[11]: import numpy as np

def flipn(p,n):
    if(n>0):
        return np.random.choice([True,False],n,p=[p,1-p])

outcome = flipn(0.67,1000)
print(outcome)

count=0
for flip in outcome:
    if flip == False:
        count+=1

print("There are " + str(count)+" tails")
```

```
[False True True True False False True False True False True True
False True True False False True True True True True False True
True True True True True False True True True True True True
False True True False True False False False True True False True]
```

[illegible]

```

False True True False True False False False False True True True
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True False True False True True False True True False True True
True True True True]

```

There are 331 tails

There are 331 tails

Question 4 4a) The discrimination problem the authors are concerned with is about discerning which author published what passages in the Federalist paper. A general definition of discrimination is to distinguish one from another.

4b) functional words are filler words such as articles, conjunctions, and prepositions. Contextual words usually only appear under certain contexts and would be dangerous to use as a discriminator. For instance, 'war' appear frequently in both Madison and Hamilton's papers when addressing armed forces.

4c) As I stated in the previous question, contextual words could be dangerous to use as a discriminator because their frequency varies greatly depending on the context of the paper, but functional words have relatively stable frequencies, which could be a good discriminator.

Question 5

```
[14]: import pandas as pd
```

```
df = pd.read_csv('shared/data/federalist-papers/function_word_counts.csv')
print(df.head())
```

	a	all	also	an	and	any	are	as	at	be	...	who	will	with	would	\
0	35	3	2.0	3	37	7.0	8	16	7	46	...	0.0	10	11	8	
1	24	5	0.0	8	46	3.0	7	15	5	18	...	6.0	3	12	5	
2	35	10	1.0	9	38	7.0	15	36	2	40	...	6.0	15	12	6	
3	64	3	0.0	10	49	6.0	8	18	4	40	...	5.0	6	15	12	
4	51	14	1.0	9	47	5.0	7	21	5	52	...	5.0	37	13	12	

	your	Essay	Author	Publication	Date	total_word_count
0	0.0	Essay 52	Unknown	New York Packet	1788-02-08	1841.0
1	0.0	Essay 67	Hamilton	New York Packet	1788-03-11	1637.0
2	0.0	Essay 54	Unknown	New York Packet	1788-02-12	1994.0
3	0.0	Essay 21	Hamilton	Independent Journal	NaN	1992.0
4	0.0	Essay 58	Madison	Unknown	NaN	2083.0

[5 rows x 75 columns]

```
[19]: print(df['Author'].value_counts())
```

```
Hamilton      51
Madison       15
Unknown       11
Jay           5
Hamilton and Madison  3
Name: Author, dtype: int64
```

```
[20]: print(df['shall'].mean())
```

2.988235294117647

```
[9]: import pandas as pd
```

```
df = pd.read_csv('shared/data/federalist-papers/function_word_counts.csv')
df_two_columns = df[['Author', 'when']]
df_two_columns.set_index('Author')
```

```
[9]:
```

	when
Author	
Unknown	3.0
Hamilton	0.0
Unknown	5.0
Hamilton	2.0
Madison	2.0
...	...

```

Hamilton    0.0
Hamilton    5.0
Madison      0.0
Madison      1.0
Hamilton     1.0

```

[85 rows x 1 columns]

```
[10]: df_two_columns.set_index('Author').idxmax(axis=0)
```

```
[10]: when      Hamilton
      dtype: object
```

```
[14]: df.groupby('Author').sum()
```

```
[14]:
```

	a	all	also	an	and	any	are	as	at	be	\
Author											
Hamilton	2478	431	33.0	632	2679	364.0	572	956	376	2274	
Hamilton and Madison	92	20	2.0	19	215	8.0	32	36	17	27	
Jay	100	24	11.0	17	408	22.0	53	102	22	160	
Madison	819	162	31.0	174	1210	86.0	215	393	82	806	
Unknown	477	87	17.0	81	545	62.0	149	220	70	532	

	...	were	what	when	which	who	will	with	\
Author	...								
Hamilton	...	148.0	145.0	109.0	1231	256.0	686	608	
Hamilton and Madison	...	21.0	2.0	8.0	56	20.0	6	41	
Jay	...	15.0	11.0	15.0	56	35.0	72	60	
Madison	...	76.0	43.0	22.0	451	78.0	284	225	
Unknown	...	42.0	30.0	22.0	260	40.0	190	115	

	would	your	total_word_count
Author			
Hamilton	922	16.0	111035.0
Hamilton and Madison	11	0.0	5600.0
Jay	68	3.0	8373.0
Madison	171	5.0	40772.0
Unknown	98	0.0	21906.0

[5 rows x 71 columns]

```
[ ]:
```